EveSense

Eavesdropper detection on a simulated BB84 link that is already noisy

Team ChiaraDev | Chiara De Venuto | Qiskit Fall Fest 2026 @ MPSTME, IBM Quantum

Problem addressed

BB84 gets its security from physics: measuring a qubit in the wrong basis disturbs it. An intercept-resend attacker inserts errors, and the two ends read those errors as a rising quantum bit error rate (QBER). Above roughly 11% QBER the link stops producing usable key.

Deployed fiber is never clean. Raman scattering, detector dark counts and polarization drift put real links at a few percent QBER with nobody listening. So the receiver is not comparing a measurement against zero. It is comparing against a baseline that drifts, and it has to decide whether a rise in QBER is an attacker or the weather. That is a detection problem with a hit rate and a false alarm rate, and this project treats it that way.

Quantum approach and Qiskit implementation

One BB84 round is one qubit. Alice prepares |0>, |1>, |+> or |-> from her bit and her basis choice. Eve, when active, intercepts, measures in her own basis and resends the state her measurement implies. The channel adds depolarizing noise on the gates and readout noise on the detectors. Bob measures in the basis he picked before the round started.

Rounds are packed into batches of 64 qubits inside a single circuit, so 30000 rounds cost milliseconds. Aer's stabilizer method handles the whole circuit, which is legitimate here because every gate used is Clifford. Eve's measurement outcome is drawn from the Born rule, which is exactly what a mid-circuit measure-and-resend produces.

Built on Qiskit 2.5.2 and Qiskit Aer 0.17.2. Seeded throughout, so every figure in the repository reproduces exactly.

Experiments

Experiment	What it varies	Rounds
Interception sweep	Eve's interception probability, 0 to 1, clean channel	30000
Channel noise sweep	Depolarizing probability, 0 to 20%, no attacker	30000
Detection study	9 conditions, 12 independent runs each	1.3 million
Key rate curve	QBER against secure key rate	closed form

Results

On a clean link the simulated QBER tracks the analytical prediction of 25% times the interception probability. The measured value at full interception is 0.2523 against a theoretical 0.25, which is the sanity check a QKD implementation has to pass before anything else it reports means anything.

Channel noise moves the same metric by the same order of magnitude. A 7% depolarizing channel produces about 7% QBER with nobody listening, and by 12% the link crosses the 11% secrecy limit on its own.

The detection study is where the result lives. On a 7% noisy link with no attacker we measure 8.3% QBER with a standard deviation of 0.9. With the same link tapped at 10% interception probability we measure 10.3% with a standard deviation of 0.5. The distributions touch. A receiver with one threshold cannot separate them without paying either missed detections or false alarms, and the ROC curves show exactly that tradeoff: on clean fiber the detector separates tapped from untapped at almost any threshold, on dirty fiber the curve collapses toward the diagonal at low interception rates.

Analysis and impact

The numbers point at a tradeoff rather than a fix. A quiet attacker stays under the noise floor, and no threshold on a single QBER measurement separates her from a bad day on the fiber. This is why real QKD deployments spend their budget making the link clean instead of making the detector clever: the security proof bounds the attacker, not the weather.

Limitations and future scope

Eve here is a textbook intercept-resend attacker. Photon-number-splitting and coherent attacks are not modeled, and nothing in this work is a security proof. The channel is depolarizing plus symmetric readout noise, while real fiber drifts in polarization and loses differently per wavelength. The next version would estimate the baseline online and run detection on the residual error, which is the version that would survive contact with a real link.

This document was developed with the assistance of generative AI tools for research, structuring, and content generation. Final content has been reviewed and validated by the author. Full compliance with: Reg. (EU) 2024/1689 (AI Act) Art. 50 | Italian Law 132/2025 Art. 13 | EU Code of Practice on AI Transparency (2026) | U.S. AI Labeling Act (S. 4915) | U.S. H.R.8479 | U.S. Executive Order on AI (Dec 2025) | California TFAIA & AB 2013 (eff. Jan 2026).